Author-prepared AI assistance declaration

A sixteen-prime-factor bound for Lehmer's totient problem and exact prime-completion criteria

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Anthropic Claude through Claude Code assisted the earlier computational development, drafting and partial Lean work. OpenAI Codex assisted later ceiling and one-candidate reductions, independent primality certificates, finite reproduction checks, the fixed-count construction, and editing and packaging. The present revision adds Codex-assisted exact derivations, descent-obstruction analysis and diagrams in the final-prime appendix.

Python was used for exact arithmetic and certificate checks; the new pivot-lemma audit uses integer and rational arithmetic for its decisive inequalities. Illustrative decimal logarithms are not proof assumptions. LaTeX and pdfLaTeX were used to prepare and compile the documents.

The historical archive contains partial Lean results. The version 1.8.0 record reports ten additional checked statements in `lean/UniformFinite.lean`. No full formalisation of the search programs or the full conjecture is claimed. The present final-prime appendix is not represented as newly Lean-verified.

The text combines human-written and AI-generated material. The earlier all-parts human-understanding declaration concerned the earlier version. Author review and a new declaration are required for the added material. The author remains responsible for the final submission and citations.

This document is an unsigned author-prepared disclosure, not an official AMS form. Any declaration required by the journal's live submission portal must be completed by the author. The full Lehmer conjecture remains unproved.